

Learning to Create with AI

Disciplines for Deep Learning, Critical Inquiry, and Creative Thought

A companion paper to Learning with AI

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Abstract

Generative AI can improve a creative artifact without increasing the learner's independent creative capability. This paper develops an evidence-informed instructional framework for using AI to cultivate what I call *learned creativity*: creative capability that persists beyond the same assistance. The framework combines six disciplines: generate before consulting AI; reconstruct and translate ideas; use AI to widen rather than close the search; trace and verify load-bearing claims; build adversarial challenge into the process; and transfer what survives to new problems. A cross-cutting navigation layer addresses a common research difficulty: deciding what to do next when several plausible directions compete. These practices extend the Learning Spiral in *Learning with AI*, a cycle of initial attempt, targeted AI help, independent work, verification, and reflection, and define complementary roles for the student, AI mentor or collaborator, peers, and instructor. Recent research supports several component mechanisms and some sequencing choices, especially protecting an independent first move, structuring human agency, and returning to unaided transfer. The literature reviewed here does not test the integrated framework as a single intervention, so it is presented as a design synthesis and research hypothesis rather than a validated curriculum. For students, the framework turns AI use into repeated practice in generation, judgment, verification, and research navigation. For instructors, it provides assignment and assessment designs that distinguish what students can produce with AI from what they can later explain, adapt, or create without the same assistance.

Keywords: generative AI, creativity, deep learning, critical inquiry, human-AI collaboration, mentoring, Learning Spiral, research navigation, next-move judgment, verification, higher education

1 Introduction

Generative AI changes the economics of intellectual work. It can produce examples, analogies, objections, drafts, code, candidate explanations, alternative designs, and plausible solutions in seconds. For a student, that creates an obvious opportunity: more ideas become available at lower cost. It also creates a less obvious educational risk. If the machine performs the generation, comparison, and repair that the student needed to practice, the finished product can improve while the learner does not.

That distinction between *performance* and *learning* is central. In a 2026 experiment with university students, free access to ChatGPT improved creative performance on an assisted task, but the advantage disappeared on a later task completed without AI. Students who instead followed a regulated sequence, generating their own ideas first and then using ChatGPT to improve, develop, and evaluate those ideas, performed better on the later independent creativity task than both the free-AI and human-only groups [34].

The finding fits a broader pattern. AI can amplify creative output, but the size and durability of the benefit depend on how the human participates [4, 11, 13, 23].

The educational objective should therefore be stated carefully. The goal is not to maximize the creativity of the artifact produced while AI is present. The goal is to develop a student who can think more deeply, ask better questions, generate more original possibilities, judge those possibilities more accurately, and carry those abilities into settings where the tool is absent or where its suggestions are wrong. I use the term **learned creativity** for this retained capability: creative judgment and generative skill that the learner can reconstruct, adapt, and transfer rather than only display while the same assistance is present.

This objective is compatible with the framework in *Learning with AI* [14]. That framework distinguishes three dimensions of student development: **Core Competence**, what the student can do independently; **AI-assisted Practice**, what the student can do responsibly with AI; and **Trustworthiness**, whether the student can verify, disclose, and defend the resulting work. It also organizes AI-assisted learning through the **Learning Spiral**: frame the question, try to solve, ask for help, work independently, verify, and reflect. The present paper does not introduce a second framework. It applies that existing one to the narrower problem of cultivating inventive and creative thought.

The argument is also narrower than a claim that AI makes people creative. Creativity depends on domain knowledge, motivation, experience, opportunity, and social context, and it cannot be reduced to a prompt technique. The framework therefore assumes a competence floor: creative leaps still depend on enough technical and domain knowledge to recognize constraints, judge unusual ideas, and distinguish a promising deviation from a fluent mistake. AI can widen the search, but Core Competence remains the basis for taste, verification, and defensible selection. The studies cited here also operationalize creativity in different ways, including divergent-thinking measures, originality and flexibility scores, product ratings, and later transfer, so their outcomes should not be treated as interchangeable. The evidence is also young. A 2026 meta-analysis of 32 quasi-experimental studies reported a moderate positive average effect of generative-AI-assisted learning on student creativity, but also found substantial moderation by discipline, activity type, duration, and collaboration design [4]. A systematic review of 67 higher-education studies similarly concluded that ChatGPT supported critical and creative thinking most reliably when embedded in inquiry-oriented, scaffolded designs, while unstructured use was more often associated with cognitive offloading and weaker critical engagement [23]. A 2026 review focused on 57 STEAM studies reached a similar conditional conclusion: benefits varied with tool type, task complexity, learner characteristics, and instructional design [10]. These findings support design principles, not a universal recipe.

The paper develops six such principles. They are modern versions of older intellectual disciplines, adapted to an environment in which a student can consult a fluent generative system at any moment. Their common purpose is to preserve human agency while exploiting what AI does well. A separate navigation layer addresses a different problem: once the student has generated, compared, and challenged several possibilities, how should the next move be chosen? That question is metacognitive rather than domain-specific, and it becomes increasingly important as AI makes plausible alternatives cheap to produce.

1.1 The design problem: amplify thought without outsourcing it

For educational purposes, creative work can be viewed as repeated movement between two broad processes. One is *divergent*: opening the search space, generating alternatives, changing representation, importing analogies, and asking what else might be possible. The other is *convergent*: testing, selecting, combining, rejecting, verifying, and deciding what is worth keeping. This division is compatible with contemporary accounts that distinguish divergent and critical-convergent processes, while the usual criterion for a creative product remains some combination of originality and effectiveness or usefulness [23, 29]. Good creative work moves repeatedly between the two.

Generative AI is unusually strong at increasing the number of available candidates. That is a genuine

advantage. It can also make those candidates feel more authoritative than they are. Fluent output encourages acceptance, and repeated exposure to AI-generated alternatives can pull different people toward similar regions of the idea space. Doshi and Hauser found that access to generative AI improved individual story ratings while reducing the collective diversity of the stories produced [7]. Later work has reported related concerns about semantic convergence, familiar solutions, and passive interaction patterns [1]. These results do not imply that AI necessarily homogenizes creative work. They show why the timing and structure of AI use matter.

The design target is therefore a particular division of labor.

	Human responsibility	Useful AI contribution
Direction	Define the question, constraints, values, and purpose; diagnose the current bottleneck; choose the next move.	Surface missing questions, alternative framings, possible constraints, and candidate next moves.
Generation	Produce an initial representation or candidate set before seeing machine suggestions.	Expand the candidate set, vary assumptions, supply analogies, and combine remote concepts.
Judgment	Decide what is relevant, original enough, feasible, ethical, and worth pursuing.	Critique, compare, stress-test, and expose neglected alternatives.
Verification	Check evidence, sources, calculations, code, assumptions, and boundary cases.	Suggest tests, counterexamples, search terms, and independent derivations.
Ownership	Integrate the final reasoning and be able to explain it without the tool.	Provide iterative feedback and rehearsal.

The table is intentionally asymmetric. AI can participate in every stage, but responsibility for direction, judgment, verification, and ownership remains with the learner. That is what turns collaboration into learning rather than substitution.

1.2 Intellectual precedents

The disciplines developed below have older precedents. They emerged in different cultures for different purposes, so they should not be treated as one hidden historical curriculum. Their value here is more modest. Each tradition solved a recurring intellectual problem in a way that remains recognizable today.

Vedic recitation used multiple transformed recitation patterns to protect oral texts against corruption, demonstrating the value of knowledge that survives changes of form [16, 32]. Zhu Xi’s reading discipline emphasized receptive understanding before premature evaluation [6]. Islamic scholarship developed *isnad* and *ijazah* traditions that made chains of transmission visible and inspectable [31]. Medieval scholastic disputation required objections to be stated before a response, institutionalizing adversarial examination [19]. Havruta study made reasoning dialogic, with partners supporting and challenging one another rather than studying only in isolation [17]. The Hungarian mathematical problem culture, including KöMaL, gave students sustained exposure to difficult problems whose solution paths were not supplied in advance [20].

A contemporary account brought these traditions together as a curriculum for mastery and inventive thinking [30]. The historical causal claims should be treated cautiously. The traditions are better understood as intellectual precedents than as proof that a particular method produced exceptional thinkers. Their modern

relevance rests on the functions they perform: transformation, fair representation, provenance, opposition, dialogue, and problem solving.

Table 1: Historical precedents and their AI-era educational functions. The mapping is functional, not a claim of direct historical continuity.

Precedent	Intellectual discipline	AI-era interpretation
Vedic recitation	Transform knowledge without losing structure	Reconstruct an AI-assisted idea in a different representation and from memory.
Zhu Xi	Understand before judging	State the strongest faithful version of an AI or human argument before criticizing it.
Isnad / ijazah	Keep provenance visible	Trace important claims to primary evidence and distinguish source from repetition.
Quaestio	State objections before defending	Ask AI and peers to build the strongest case against the current approach.
Havruta	Think with an independent partner	Use peers, mentors, and AI as distinct minds whose disagreement is informative.
Hungarian problem culture	Learn through difficult unsolved work	Preserve periods of unaided struggle and end with transfer or problem finding.

The important change in the AI era is that challenge, explanation, and alternative generation are now almost frictionless to obtain. The educational problem is no longer scarcity of assistance. It is regulating assistance so that the learner still performs the work that builds durable capability.

2 Six disciplines for creative learning with AI

The six disciplines below regulate where AI enters creative work and which parts of the intellectual process remain the learner’s responsibility. They are not a rigid sequence. A student may move among them repeatedly as a problem develops. The navigation layer is cross-cutting: it is invoked when several plausible next moves compete and the learner needs to decide where to direct effort.

2.1 Discipline 1: Generate before you consult

The first discipline is simple: **produce something before asking AI to produce something for you**. The object can be small. It may be three candidate ideas, a diagram, a conjecture, a paragraph, a proof sketch, a list of assumptions, a design constraint, a hypothesis, or a description of where the learner is stuck. What matters is that the student’s search begins before the model’s search space becomes visible.

This practice is the creative analogue of the Learning Spiral’s *try to solve* step. It protects what the book calls the friction wall: the point where useful difficulty can either become learning or be bypassed by asking for a finished answer. Learning research has long shown that retrieval and generation can improve later retention even when the initial attempt is unsuccessful [3, 21, 27]. The newer creativity evidence now gives the same sequencing principle direct relevance to generative AI. Wong and Qiu’s regulated-AI condition required independent idea generation before AI collaboration and produced stronger later independent creativity [34].

The advantage is not merely that the student “thinks first.” Independent generation creates a baseline. It records the learner’s own prior before the model changes it. That baseline makes later reflection possible: Which ideas came from the student? Which were introduced by the model? Which changed because of critique? Which survived because they were actually good rather than merely fluent?

It also preserves diversity across a class. If thirty students begin by asking the same model the same broad question, they are likely to sample from overlapping regions of its response distribution. If they begin with thirty independently generated candidate sets, AI enters thirty different trajectories. The point is not to keep AI from influencing the work. It is to delay that influence until the learner has supplied a direction of their own.

The main disadvantage is time. Independent attempts can feel inefficient, particularly when a student knows that AI can immediately provide competent options. A novice may also lack enough domain knowledge to generate useful candidates. The remedy is not to abandon the discipline but to calibrate its scale. For a novice, the first attempt might be a list of what is known, what is confusing, and one tentative idea. For an advanced student, it can be a full independent approach.

A productive student prompt begins only after that record exists:

I generated the three approaches below before asking for help. Do not replace them yet. For each one, identify what assumption it relies on, then propose one substantially different approach that starts from a different assumption. Tell me what the alternative reveals about my original idea.

The prompt uses AI for divergence while keeping the student's first move visible.

2.2 Discipline 2: Represent, reconstruct, and translate

A creative mind needs more than a large idea supply. It needs representations that can be moved, recombined, and inspected. The second discipline therefore has two parts: represent an idea fairly before changing it, then reconstruct it in another form without leaning on the original explanation.

The first part protects against premature critique. Before asking whether a theory, argument, design, or AI response is wrong, the student should be able to state what it actually says. This is especially important with generative AI because the tool makes rebuttal cheap. A student can produce ten objections to a position they have not yet understood. Requiring a faithful representation prevents the appearance of critical thinking from replacing comprehension.

The second part tests whether the idea has become portable. The *Learning with AI* framework calls this the translation test: can the learner express the same structure in another modality, notation, example, diagram, explanation, or derivation [14]? Translation is useful for creativity because ideas become easier to combine when they are represented at more than one level. A formula can become a graph; a graph can become a physical story; an argument can become a dependency map; a design principle can become a counterexample; code can become an algorithm described in plain language.

AI is valuable here when it acts as a mentor rather than a replacement. It can ask the learner to translate, compare the learner's reconstruction with the original, or provide a new representation after the learner has tried. It can also help move between levels of abstraction. The danger is that an AI-generated explanation can produce a strong feeling of understanding while leaving the learner unable to reproduce the idea. The answer is a blank-page return.

A simple protocol is:

1. Read or discuss the idea until it can be stated fairly.
2. Close the source and the AI window.
3. Reconstruct the idea from memory.
4. Change representation: diagram, example, notation, analogy, algorithm, or dependency graph.
5. Reopen the source and compare. Mark what was lost or distorted.

This sequence draws on retrieval practice, generation, and self-explanation. These mechanisms have a longer evidence base than generative AI itself [8, 27]. Its disadvantage is that repeated reconstruction can become mechanical. The instructor should therefore vary the transformation and ask whether the new representation exposes something that the original concealed.

For creative work, the most useful transformation is often not “say it again differently” but “make it answer a different question.” Ask how the idea changes if a constraint is removed, if the scale changes, if the audience changes, if a variable becomes stochastic, if the mechanism is reversed, or if the design must work under a hostile boundary case. Representation then becomes a route to invention.

2.3 Discipline 3: Use AI to widen the search space, not to choose for you

Generative AI’s clearest creative advantage is breadth. It can rapidly produce analogies, alternative hypotheses, examples from other domains, unusual combinations, candidate mechanisms, and competing designs. The third discipline is to use that breadth deliberately while reserving selection for the learner.

The distinction matters because unstructured brainstorming with AI can become passive. Studies of human-AI creative interaction have found that students often use directive prompts and repeat or lightly edit AI suggestions rather than engaging in deeper combination and transformation [1]. By contrast, collaborative interaction that improves and develops self-generated ideas is associated with stronger later independent originality [34]. Human agency appears to be a central moderator: deep integration of human ideas with AI output is more promising than either AI dominance or a token independent phase added only at the beginning or end [11]. The evidence is not uniformly positive. In a separate experiment, human- and ChatGPT-provided primes produced no general difference in divergent-thinking scores, while a low-creativity prime attributed to ChatGPT reduced performance in one condition [25]. That result reinforces the need to treat AI output as material for judgment rather than as creativity transferred automatically to the learner.

A useful discipline is to require *structured divergence*. Instead of asking “give me ten ideas,” ask the model to traverse different regions of the space. For example:

Here is my current approach. Generate four alternatives under four different constraints: one that minimizes cost, one that maximizes robustness, one that assumes my main mechanism is wrong, and one imported from a distant field. Do not rank them. For each, state what new possibility it exposes and suggest one low-cost prototype or test that would make the alternative concrete. Do not choose among them.

This prompt makes diversity an explicit target. Other useful variation operators include changing scale, time horizon, stakeholder, objective function, representation, causal mechanism, data source, and failure assumption. One especially valuable operator is **orthogonal search**: when the current representation stalls, deliberately ask what another field, scale, data source, or method would make visible. Hassabis describes using neuroscience as an orthogonal source of information after reaching the limits of the AI approaches then available to him; the example is biographical rather than experimental evidence, but the heuristic is broadly useful for research navigation [12].

AI also changes the cost of making alternatives concrete. A useful rule is **prototype broadly; commit late**. After an independent first move, the learner can use AI to turn several competing possibilities into sketches, toy models, short simulations, code fragments, storyboards, experimental designs, or other inexpensive probes before investing heavily in one direction. Hassabis reports seeing expert artists use AI in this way: rapid prototyping lets them examine more candidate ideas before committing scarce time and resources [12]. This is expert observation, not causal evidence that prototyping with AI increases learned creativity. Educationally, the value lies in making alternatives inspectable while leaving the criteria and final commitment with the learner.

The student should not ask AI to select the “best” idea immediately. Early ranking encourages convergence before the search space has been explored. A better sequence is generate, cluster, compare, combine, and only then evaluate. The human should be able to explain the criteria by which alternatives are rejected. In an **AIC** assignment, AI can contribute directly to the candidate set. In an **AIT** assignment, the AI should instead ask questions that help the student generate the alternatives.

The instructor has a corresponding responsibility: grade the reasoning behind selection, not the number or polish of ideas. If only the final artifact is graded, students are rewarded for outsourcing divergence and convergence together. Process evidence can be light: a short record of the student’s pre-AI ideas, the alternatives explored with AI, and the reasons the final direction was chosen.

There is also a class-level reason to preserve this record. AI can raise individual creative performance while narrowing collective variation [7]. An instructor can monitor whether project topics, examples, or solution strategies are converging across a cohort. If they are, the response should usually be to diversify starting data, constraints, source sets, or required perspectives rather than simply prohibit AI.

2.4 Discipline 4: Trace claims and verify the load-bearing parts

Creativity without verification can produce persuasive nonsense. The fourth discipline makes provenance and checking part of creative work rather than a final compliance step.

The historical chain-of-custody idea translates naturally to AI-assisted inquiry: a claim should not travel alone. Important claims should carry a route back to evidence that the learner has inspected. AI makes this more important because it can generate citations, summaries, factual statements, calculations, and code whose fluency exceeds their reliability. In the Learning Spiral, verification is the step that turns assistance into responsible academic work [14].

Verification is not the same as asking the same AI whether it is correct. The check should be independent in the sense appropriate to the field. For a factual claim, open the primary or authoritative source. For mathematics, derive the result independently, test limiting cases, or seek a counterexample. For code, run known-input tests and boundary cases. For data analysis, reproduce the result with an independent implementation or benchmark. For an argument, identify what evidence would change the conclusion.

The creative value of verification is easy to underestimate. A failed check is not only error detection. It often identifies the place where a new question begins. A contradiction can expose an unstated assumption. A source that does not support a repeated claim can reveal an open empirical gap. A counterexample can suggest the right theorem. Verification therefore belongs inside the creative process.

AI can assist without being the final judge. It can propose tests, generate adversarial examples, list assumptions, identify primary-source candidates, or suggest independent derivations. The student then performs the decisive check.

The main disadvantage is cost. Verification slows the work and can feel like a tax on ideation. The remedy is to verify the *load-bearing* parts first. Not every sentence deserves the same effort. Students should identify the claims on which the conclusion would fail if they were wrong and spend verification effort there.

This principle also helps instructors. An assignment can ask students to flag one AI contribution they rejected after verification and explain the check that exposed the problem. Such an artifact is more informative than a generic statement that “I verified the output.” It provides evidence of judgment.

2.5 Discipline 5: Build an adversary into the process

Creative people need encouragement, but ideas improve through resistance. The fifth discipline is to create structured opposition before commitment becomes expensive.

AI seems well suited to this role because it is always available and can take many positions. Yet this is also where naive use is risky. Conversational systems can be sycophantic, shaping responses toward what the

user appears to want. Recent experimental evidence shows that sycophantic AI can increase dependence and influence behavior even when users evaluate the AI source less favorably [5]. A prompt that asks “Is my idea good?” therefore creates the wrong incentive. It invites affirmation.

The alternative is to specify the adversarial task. Ask for objections before praise. Ask for counterexamples, hidden assumptions, alternative causal stories, missing evidence, and failure conditions. Separate the roles of advocate and critic. Better still, use more than one independent critic: a peer, an instructor, and an AI model do not share the same priors or failure modes.

Treat the following idea as a claim you are trying to break. Give me the strongest objection first. Then identify one hidden assumption, one plausible counterexample, one missing piece of evidence, and one condition under which the idea would still survive. Do not rewrite the idea for me.

This is the modern counterpart of structured disputation. Its educational purpose is not contrarianism. It is to make the learner represent opposing evidence accurately enough to respond to it. A useful closing move is role reversal: after receiving the critique, the student must defend the objection as strongly as possible before returning to their original position.

The strongest evidence for this design is not limited to AI. Constructive controversy and peer instruction have long shown the value of explanation and disagreement in learning [24]. Recent AI-mediated work adds a relevant result. A critical-thinking intervention combining authentic instruction, structured dialogue, and AI-supported peer mentoring reduced direct adoption of AI content and produced more original solutions than unstructured use [13]. The short intervention did not significantly change self-reported critical-thinking scores. That distinction matters: behavior changed before the self-report did.

The main failure mode is performative opposition. An AI can produce shallow objections on command, and students can learn to answer them formulaically. Instructors should therefore occasionally supply objections that the AI did not generate, or ask students to compare objections from different sources and identify which one genuinely threatens the argument.

2.6 A cross-cutting navigation layer: choose the next high-value move

The six disciplines describe how to generate, understand, widen, test, challenge, and transfer ideas. Complex inquiry adds another problem that cuts across all six: the learner may have several plausible directions and no obvious reason to prefer one. At that point the main difficulty is not object-level reasoning, solving the problem itself, but meta-level reasoning about where attention should go next.

This is a form of **next-move judgment**. The aim is not to discover a uniquely correct next step. Research rarely offers one. The aim is to choose the most informative move given the present state of knowledge, execute it, and then diagnose the problem again. The decision is therefore provisional and iterative.

Recent work on human–AI creativity gives this metacognitive layer direct empirical relevance. In a 2026 mixed-methods study of university students, initiative and metacognitive integration were strongly associated with pathways leading to emergent creativity [15]. Because the quantitative component used a structural model rather than a randomized intervention, it does not establish that teaching next-move judgment causes creativity gains; it does support treating metacognitive monitoring and control as explicit parts of human–AI creative work.

A useful navigation routine has six parts.

1. **Restate the current question.** Say what you are trying to understand, establish, explain, design, or create now, not what the project was about three steps ago.
2. **Locate the bottleneck.** Identify what is currently preventing progress: missing knowledge, a doubtful assumption, contradictory evidence, an untested computation, too many alternatives, or failure of the current mechanism.

3. **Scan for anomalies.** Look for what is surprising, inconsistent, conspicuously absent, or stronger than the evidence warrants. An anomaly is often more informative than another routine check because it points to where the current mental model is under strain.
4. **Generate a small candidate set.** Name at most three plausible next moves: derive a special case, search for a counterexample, inspect a primary source, run an experiment, change representation, test an assumption, run an orthogonal search from another field or method, or pursue an alternative mechanism.
5. **Compare the candidates counterfactually.** For each, ask what you would learn if it succeeds, what you would learn if it fails, whether failure would be informative by ruling out a mechanism or exposing an assumption, whether it could invalidate the current direction, whether it unlocks other questions, how much effort it requires, and whether the result can be checked independently.
6. **Choose one move and state why.** Commit only to the next iteration. After carrying it out, reassess rather than continuing automatically down the same path.

This comparison treats **informative failure** as a positive research outcome. A high-value move need not be the one most likely to succeed; it may be the one for which either outcome changes what is known. Hassabis describes this research posture as pursuing directions that, if they fail, fail in a way that still adds knowledge [12]. The principle is especially useful when AI generates many plausible routes: probability of success alone is a poor proxy for information value.

Two principles make the routine useful. First, *blockers come before optimizations*. If a result may be mathematically wrong, a source does not support a load-bearing claim, or a key assumption is untested, polishing exposition is not the high-value move. Second, *the bottleneck matters more than the most visible imperfection*. AI is especially useful for proposing competing diagnoses because it can inspect the problem from several perspectives quickly. It should not own the final decision.

I am uncertain what to do next. Do not solve the problem for me. First help me diagnose where progress is blocked. Ask what I am trying to accomplish, what I have tried, what evidence I have, and what remains uncertain. Then identify at most three plausible next moves. For each, explain what I would learn if it succeeds or fails, its likely effort, and whether it could change the direction of the project. Recommend one move, but make me decide whether I agree and explain my reasoning before helping me carry it out.

The routine has a boundary condition. Excessive optimization can suppress serendipity. A student who treats every exploratory branch as an efficiency calculation may reject strange or difficult ideas too early. The navigation layer therefore chooses *one move for now*, not one path forever. It is a way to recover direction when inquiry branches, not a rule against wandering.

This layer belongs between passes through the Learning Spiral rather than as a seventh discipline. Pólya's classic problem-solving method already treats looking back and asking what a solution opens as part of mathematical work [26]. Generative AI makes the navigation problem more visible because it can produce many credible alternatives in seconds. The educational task is to make the learner better at selecting among them without surrendering that selection to the tool.

2.7 Discipline 6: Integrate, transfer, and find the next problem

The final discipline asks whether the student can do something new after the collaboration. This is where assisted performance becomes either learning or dependence.

After AI has helped with an idea, the student should close the tool and produce the next step independently. That step might be a new proof, a variant design, a second case, a new dataset, an extension of the argument, or a new problem formulated from the limitations of the first. The test is not whether the student remembers the AI conversation. It is whether the underlying structure can travel.

This is the creative version of transfer. It is also the most important place to preserve human agency. Guo and colleagues found that simply placing independent thinking before or after AI collaboration was not sufficient to sustain agency across the process; deeper integration of human ideas with AI output was associated with stronger agency and creativity [11]. The implication is that independence should recur throughout the interaction, not appear as a ceremonial first or last step.

A practical pattern is:

1. Generate independently.
2. Collaborate with AI to widen, compare, or critique.
3. Close AI and integrate the useful pieces in the student's own structure.
4. Reopen AI only for a targeted challenge or missing piece.
5. Close AI again and complete a transfer task.
6. Return after a delay and reconstruct the central insight without assistance.

The last two steps matter because immediate assisted success can hide a recognition-production gap. Spaced retrieval and mixed practice are well-supported methods for durable learning [8, 28]. In a creativity course or project, delayed return can be more ambitious than recall: ask the student to formulate the next question. What assumption now looks fragile? What nearby case is unsolved? What result would be surprising? What constraint, if reversed, creates a new design problem?

This is where creativity and research begin to overlap. The learner is no longer only solving assigned problems. They are learning to notice which problems are worth asking.

3 From disciplines to practice: the Learning Spiral

The six disciplines fit naturally inside the Learning Spiral rather than beside it. The center remains the question. The difference is that a creative pass through the Spiral does not aim only at a correct answer. It aims to change the question, the representation, or the space of possible answers.

The navigation layer does not change the five canonical steps. When reflection makes the next question obvious, the learner simply begins another pass. When several next questions or actions compete, the Next-Move Check sits between passes: diagnose the bottleneck, compare a few candidate moves, choose one provisionally, and then re-enter the Spiral. In this sense, navigation is a control layer for the Spiral rather than another point on it.

The Spiral also clarifies the difference between **AIT** and **AIC** work. In an **AIT** assignment, AI helps the student learn but does not contribute material that becomes part of the submitted answer. The most useful creative roles are questioning, critique, hinting, analogy, and rehearsal. In an **AIC** assignment, AI may contribute directly to the final product. The student then needs stronger process evidence showing how alternatives were generated, what was selected, what was rejected, what was verified, and what the student can defend independently.

The navigation layer is intentionally absent from table 2. It is not a seventh kind of intellectual work. It is the metacognitive routine used when the learner must decide which kind of work is most valuable next.

3.1 For students: learning and creating with an AI mentor

Students need a practical routine that is demanding enough to preserve learning but light enough to use repeatedly. The following sequence is a creativity-oriented implementation of the Learning Spiral.

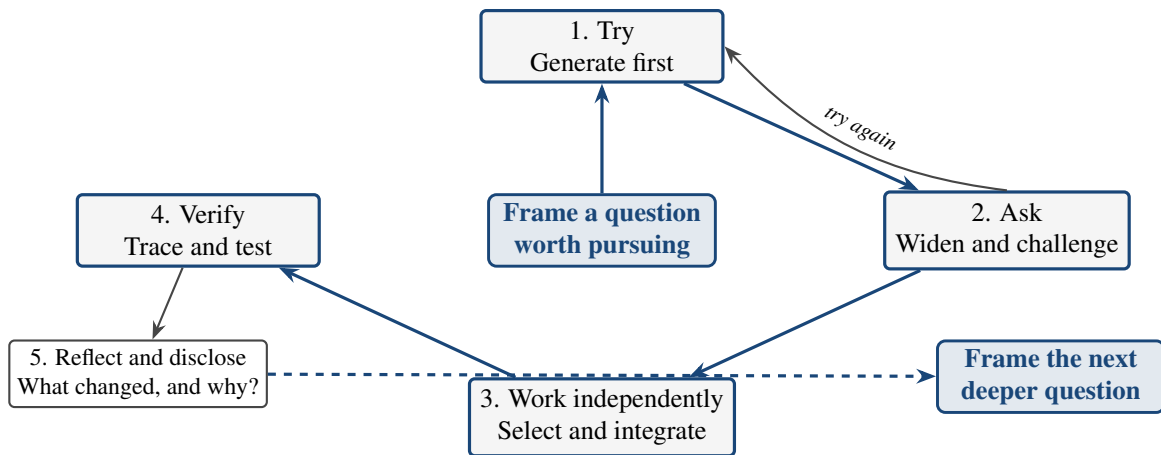


Figure 1: **The Learning Spiral applied to creative inquiry.** The structure is the same Learning Spiral used in *Learning with AI*; the annotations make the creative function of each step explicit. AI enters after an initial human attempt, leaves again during independent integration and transfer, and can re-enter for targeted challenge.

3.1.1 Start with a private first move

Before opening AI, write the question in your own words and produce a first move. For creative work, a first move should usually contain more than one candidate. Three rough ideas are often better than one polished idea because they make comparison possible. Keep this first record. It becomes the reference point for the rest of the session.

3.1.2 Tell the AI how to help

An AI mentor should not infer that every request is a request for an answer. State the role explicitly. Ask it to question, compare, challenge, or broaden. The configuration approach in *Learning with AI* is useful here: the AI should know the learner’s level, should reveal its calibration, and should reduce scaffolding as the learner demonstrates independence [14].

For creative work, add one instruction: do not collapse the search space too early. A mentor should resist giving a single polished direction when the learner is still exploring.

Act as a mentor for creative problem solving. I will show you my current ideas first. Help me widen the search space, identify assumptions, and compare alternatives. Do not choose a final answer for me. When I ask for critique, challenge the idea directly rather than reassuring me. When I appear ready to decide, ask me to state my own criteria before you rank anything.

3.1.3 Alternate collaboration with independence

Long uninterrupted AI sessions encourage cognitive offloading. Fan and colleagues describe a form of metacognitive laziness in which learners use generative AI to complete cognitive work while monitoring the task less carefully [9]. The remedy is not a fixed time limit. It is a repeated alternation: ask, close, produce; ask, close, produce.

A useful self-check is the question already used in the book: *What have I produced since the last time I asked?* If the answer is nothing, the next prompt is likely to move the student farther from ownership.

Table 2: Six AI-era disciplines inside the Learning Spiral.

Discipline	Spiral location	AI contribution	Human evidence
Generate before consulting	Frame / Try	Delayed until an initial human candidate exists	Pre-AI notes, sketch, candidate set, or attempted solution
Represent and reconstruct	Try / Work independently	Explanation, comparison, alternative representation	Blank-page reconstruction; translation to another form
Widen the search space	Ask for help	Divergent alternatives, analogies, constraint changes	Selection criteria and reasons for combining or rejecting ideas
Trace and verify	Verify	Candidate tests, sources, counterexamples, independent routes	Primary-source checks, tests, derivations, reproducible evidence
Build an adversary	Ask / Verify	Objections, failure cases, competing explanations	Steelman of the objection and reasoned response
Integrate and transfer	Work / Reflect	Targeted feedback after independent synthesis	New problem, delayed reconstruction, or unassisted transfer task

3.1.4 When you do not know what to do next

Uncertainty about the next move is normal in complex work. Do not respond by asking AI, “What should I do?” and following the first confident answer. Run the Next-Move Check instead. State the current question, name the bottleneck, look for anomalies, generate at most three candidate moves, compare what each could teach you, and choose one yourself.

The best next move is often not the move most likely to succeed. Prefer moves whose failure would still teach you something important. A counterexample search may be valuable precisely because either outcome is informative. A small experiment may be preferable to another hour of abstract argument because it discriminates between two mechanisms. A primary-source check may outrank a new idea because the entire direction rests on a claim that has not been verified.

After choosing, write one sentence explaining the decision before you act. This small commitment makes the student’s judgment visible and gives the next reflection something concrete to evaluate.

Help me run a Next-Move Check. Do not choose for me until I have stated my own diagnosis. Ask me for the current question, the main bottleneck, and any anomaly or result I cannot explain. Then help me formulate at most three next moves and compare what each would teach me if it succeeds or fails. End by asking me which move I choose and why.

3.1.5 Keep a decision record, not a transcript diary

Students do not need to document every prompt. They should record the decisions that changed the work: what they tried first, which AI suggestion altered the direction, which suggestion they rejected, what they verified, and what they can now do without AI. This is enough to support reflection, disclosure, and an instructor’s assessment of agency.

3.1.6 End beyond the original task

The session should end with a transfer move: a new example, a new constraint, a new case, a new question, or a blank-page reconstruction. A student who can only repeat the assisted solution has completed the product

but not the learning cycle.

4 Instructor design, assessment, and course implementation

Students will not reliably develop these habits simply because a syllabus says “use AI critically.” The assignment has to make the habits useful and visible. Instructor design therefore mirrors the six student disciplines.

Table 3: Student disciplines and matching instructor responsibilities.

Student discipline	Instructor design responsibility	Evidence worth collecting
Generate before consulting	Require or reward an independent first move before AI access	Initial ideas, predictions, sketch, hypothesis, or problem representation
Reconstruct and translate	Ask for another representation or an oral/blank-page explanation	Diagram, explanation, re-derivation, alternative notation, live defense
Widen deliberately	Build assignments with multiple defensible paths and varied constraints	Candidate set plus reasons for selection; diversity of approaches
Verify load-bearing claims	Make checking part of the grade, not an optional afterthought	Source trace, test case, reproducibility note, counterexample attempt
Seek adversarial challenge	Require serious objections and role reversal	Objection-response pair; peer critique; revision memo
Transfer and problem-find	Include a later or variant task where AI is absent or constrained	Unassisted transfer, delayed return, new question, or extension proposal

4.1 Use the assignment categories deliberately

The four assignment categories from *Learning with AI* are useful because creative work needs more than one relationship to AI.

- **NAI** tasks establish Core Competence and protect moments of independent generation, transfer, and performance.
- **AIT** tasks are well suited to mentoring, questioning, explanation, critique, rehearsal, and guided divergence. The student submits work that remains their own.
- **AIC** tasks are appropriate when co-creation is part of the learning goal. AI may contribute to the final artifact, so process evidence, verification, and disclosure become more important.
- **AIS** tasks make AI itself the object of inquiry. Students can compare model outputs, measure convergence, audit bias, study hallucinations, or test whether different prompting strategies change originality.

A course that aims to cultivate creativity should probably use all four at different moments. A purely **AIC** course can produce strong artifacts while leaving independent creativity unmeasured. A purely **NAI** course gives up the opportunity to teach collaboration with a tool students will encounter elsewhere.

4.2 Protect the learner’s first move

The clearest design lesson from current evidence is to protect independent generation before AI assistance. This can be done without surveillance. In class, give students five minutes to sketch candidate ideas before

opening devices. Online, ask for a short pre-AI note. In project courses, require an initial question map or proposal before a tool-assisted ideation session.

The point is not proof of innocence. It is cognitive sequencing. The instructor is creating the same regulated condition that appears promising in recent experiments [34].

4.3 Teach collaboration strategies explicitly

Students often know how to ask AI for content before they know how to collaborate with it. Instructors should model a small number of high-value moves: ask for alternatives under different assumptions, ask for counterexamples, ask for a comparison rather than a verdict, ask the model to identify what evidence would discriminate between two hypotheses, and ask it to critique without rewriting.

This is analogous to teaching laboratory technique. Access to equipment is not the same as knowing how to use it. Likewise, a general instruction to “brainstorm with ChatGPT” is weaker than a worked demonstration of how to preserve agency while using the model’s breadth.

At the course level, this suggests a useful division of labor: use AI where individualized preparation and repeated practice scale well, and protect human meeting time for activities that depend on independent minds—project work, explanation, peer challenge, collaborative judgment, mentoring, and team creativity. Hassabis has argued for a similar inversion of the classroom, with personalized AI-supported preparation paired with more project-based and collaborative in-person work [12]. His proposal is an expert design perspective rather than evidence that this arrangement is generally superior. For instructors, the practical test is whether moving a task to AI-supported preparation frees human time for cognitive work that the course actually values rather than merely moving more content outside class.

4.4 Make next-move judgment visible

Students need practice deciding where to direct effort, not only practice executing well-defined tasks. In project work, instructors can make this judgment visible with a short **Next-Move Memo**. It should be light enough to repeat:

Current question	What are we trying to establish, understand, design, or explain now?
Current bottleneck	What presently prevents progress?
Candidate moves	What are the two or three strongest things we could do next?
Chosen move	Which one will we do now, and why is it more informative than the alternatives?
Result	What happened, including failed attempts or negative evidence?
Revised question	What should the next pass through the Learning Spiral now address?

The memo is useful precisely because the final answer may not yet exist. It lets an instructor assess developing research judgment while the work is still uncertain. It also separates AI assistance from ownership: AI may suggest candidate moves or challenge the diagnosis, but the student must identify the bottleneck and defend the choice.

For mentoring conversations, four questions are often enough: *What is the most important uncertainty now? What are the strongest things you could do next? Which one did you choose? Why is it more informative than the alternatives?* Over a semester, improvement in these answers is evidence that the student is becoming less dependent on assigned pathways.

4.5 Design for disagreement among minds

AI should not replace human collaboration. Peers and instructors contribute different forms of resistance. A peer brings another human’s prior knowledge and values. An instructor can recognize disciplinary significance and detect polished nonsense that resembles competence. AI contributes availability, breadth, and rapid iteration.

The most effective setting is therefore often a small system rather than a dyad: student, AI mentor, peers, and instructor. Their roles should not be identical. A 2026 scoping review of generative AI in higher-education group work found both benefits, including idea generation, reflective support, communication efficiency, coordination, and feedback, and risks, especially reduced peer interaction and critical engagement under over-reliance. Across the reviewed studies, outcomes depended on task design, degree of AI integration, AI literacy, and subject expertise [33].

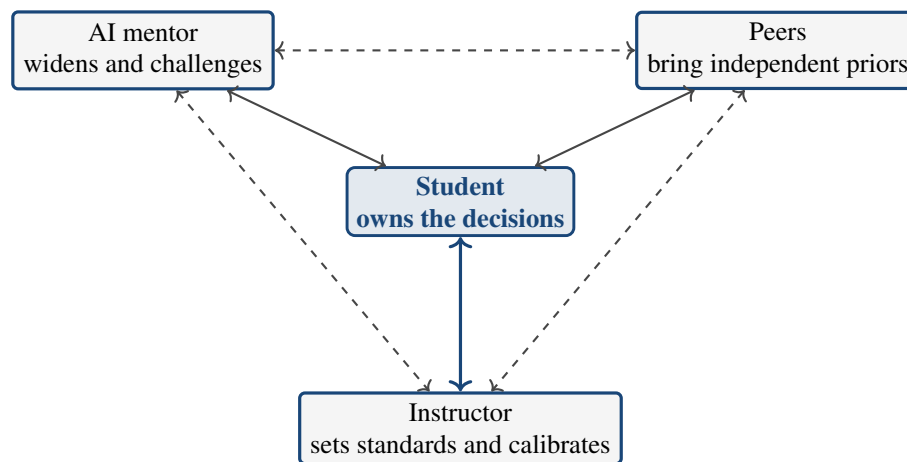


Figure 2: **A mentoring and collaborative setting for creative learning.** The student remains the locus of direction and ownership. AI supplies breadth and rapid challenge; peers provide independent human perspectives; the instructor calibrates difficulty, standards, and evidence. The dashed links indicate that peers and instructors can also interrogate the AI contribution rather than treating it as private to the student.

4.6 Do not grade polish as a proxy for creativity

AI raises surface quality quickly. An instructor who grades polish alone therefore measures the tool’s contribution as much as the student’s. Creative assessment should instead separate at least four things: originality, usefulness or fit to constraints, quality of reasoning, and ownership of the decisions. In a course where creativity itself is a learning outcome, add a fifth: transfer to a new task without the same assistance.

This distinction also addresses equity. Students differ in prior tool experience, subscriptions, time, and access. Grading the reasoning and independent transfer reduces the reward for simply having better access to the most capable model.

4.7 Assessment: distinguish assisted creativity from learned creativity

An AI-aware creativity assessment should answer two different questions.

1. **What can the student produce with responsible AI assistance?**
2. **What creative capability remains when that assistance is removed or changed?**

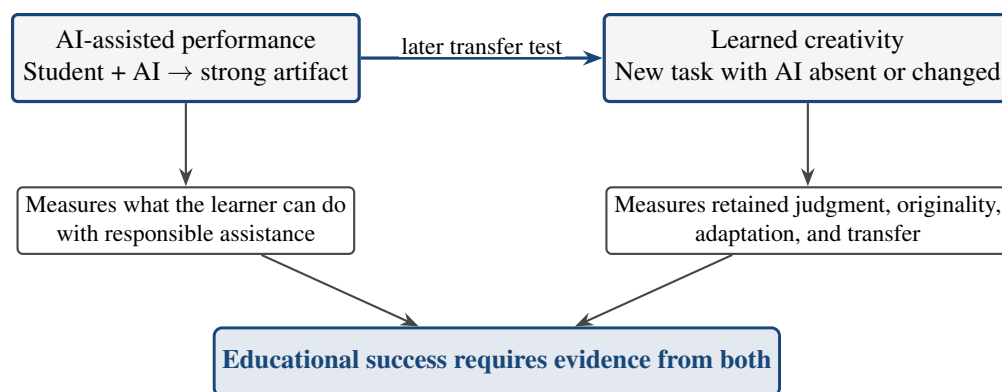


Figure 3: **Assisted creative performance is not the same as learned creativity.** An AI-assisted task measures what a student can produce in collaboration with the tool; a later task completed without the same assistance tests what judgment and creative capability the student retains and can transfer. Educational evaluation should sample both when creativity is a learning outcome.

The first is AI-assisted Practice. The second is Core Competence. Trustworthiness connects them because the student must show how the assisted work was produced and checked.

A useful assessment design pairs one assisted task with one later transfer task. The transfer task should preserve the underlying structure while changing enough surface features to prevent simple copying. This is close to the logic of the 2026 “think first, ChatGPT later” experiment, where the key result appeared on a later independent invention task rather than on the assisted task itself [34].

For project work, an oral defense or short conference can probe ownership. Ask the student to explain why one alternative was rejected, what the AI got wrong, what evidence changed the design, and what they would try next. For large courses, the same evidence can be collected through short process artifacts and a sampled no-AI transfer check.

A compact rubric might use five rows:

Criterion	What strong evidence looks like
Independent initiation	The student can show a substantive pre-AI attempt or candidate set.
Divergent exploration	Alternatives differ in assumptions or structure, not only wording.
Judgment and integration	The student can explain what was accepted, rejected, combined, and why.
Research navigation	The student can identify the current bottleneck, compare plausible next moves, and justify the move chosen.
Verification	Load-bearing claims, calculations, sources, or code were checked independently.
Transfer and ownership	The student can solve, explain, adapt, or extend the idea without reproducing AI output.

This rubric avoids an important mistake: using “number of AI prompts” as evidence of engagement. A long transcript can represent deep collaboration or passive dependence. The quality of the student’s decisions matters more than interaction volume.

4.8 A practical course pattern

An instructor can implement the approach without redesigning an entire course. One recurring project cycle is enough.

1. **Frame.** Students state the problem, audience, constraints, and what would count as a useful result.
2. **Independent generation.** Students produce an initial candidate set before AI access.
3. **AI mentoring, collaboration, or low-cost prototyping.** Depending on the assignment category, AI either coaches the student's thinking, contributes alternatives directly, or helps make several candidate directions concrete enough to compare before commitment.
4. **Independent integration.** Students close AI and select, combine, or rebuild the direction in their own structure.
5. **Adversarial review.** A peer and AI critic attack the current approach from different angles.
6. **Verification.** Students check the claims, calculations, sources, assumptions, or prototypes on which the result depends.
7. **Revision and disclosure.** Students revise and record the decisions that materially changed the work.
8. **Transfer.** A later task asks students to apply the underlying idea under a new condition with reduced or no AI assistance.
9. **Problem finding.** Students end by formulating one question that the completed work opened but did not answer.

The sequence should not be interpreted as nine bureaucratic checkpoints. The navigation layer is invoked only when the route becomes unclear: pause, diagnose the bottleneck, compare a few candidate moves, choose one, and resume the cycle. In a seminar it can unfold in conversation. In a large course it can be compressed into three artifacts: a pre-AI plan, a short decision-and-verification memo, and a later transfer task. The educational invariant is the alternation between human generation, AI-supported expansion, human judgment, independent checking, and transfer.

4.9 What instructors should look for over a semester

A single creative assignment can be gamed or completed performatively. Development appears across repeated work. Five trends are more informative than one polished product.

First, the student's independent first attempts should become more structured. Early attempts may be vague or narrow. Later ones should contain better questions, more relevant constraints, and stronger candidate mechanisms.

Second, AI interaction should become more selective. Stronger students usually need fewer broad "give me ideas" prompts and more targeted requests for critique, comparison, evidence, or a specific missing perspective.

Third, rejection quality should improve. A mature collaborator does not merely accept better outputs. They become better at explaining why plausible outputs should be discarded.

Fourth, next-move judgment should improve. Early in a project, students may ask what to do next in broad terms or choose the easiest available action. Later, they should be able to identify the real bottleneck, distinguish a blocker from an optimization, compare a few plausible investigations, and explain why one is the best move for the current iteration.

Fifth, transfer should strengthen. The student should increasingly be able to reproduce the central reasoning, adapt it, and formulate extensions without the same scaffold. That is the clearest sign that assisted creativity has become learned capability.

These trends also give instructors a practical mentoring agenda. Feedback can focus on the process that is currently weakest: question framing, divergence, selection, next-move judgment, verification, or transfer. AI can supply more practice at each stage, but the instructor remains responsible for deciding what development matters next.

5 Discussion: strengths, boundary conditions, and research agenda

5.1 What the framework adds

The framework's main contribution is to make **learned creativity**, rather than the quality of an AI-assisted artifact, the educational target. Independent generation preserves a student's initial trajectory. Reconstruction and translation test whether the student understands the structures being recombined. Deliberate divergence then uses AI to expand the search without allowing the model to choose the direction by default; orthogonal search and low-cost prototyping make remote alternatives easier to inspect before commitment.

It also joins creation to criticism. Adversarial challenge, provenance checks, and independent verification are not end-stage quality controls; they are ways to discover structure, expose boundary cases, and redirect an idea while it is still developing. The navigation layer adds a convergence mechanism when AI makes alternatives cheap: the learner identifies the current bottleneck, compares the information value of a few plausible actions, and chooses one next move without pretending that the rest of the research path is known.

Finally, the framework makes the human–AI relationship teachable and assessable. AI can serve as mentor under **AIT** or as co-producer under **AIC**, while peers and instructors supply independent judgment and disciplinary standards. The observable behaviors are concrete: try first, ask for a particular kind of help, return to independent work, verify the load-bearing parts, show what changed, and demonstrate transfer when the assistance is removed or altered.

5.2 Boundary conditions

Status of the framework: component evidence, system hypothesis. The framework proposed here has not been tested as an integrated intervention. Its individual elements draw on established learning research and a growing AI-specific literature, and some sequencing choices now have direct experimental support. None of the studies reviewed here tests the exact six disciplines, navigation layer, Learning Spiral, and assessment design as a single package. The framework should therefore be read as an evidence-informed instructional design synthesis that generates testable hypotheses, not as an empirically validated curriculum. The same caution applies to the six-step Next-Move Check: current evidence makes metacognitive monitoring and control plausible targets for instruction, but does not establish that this particular routine causes gains in creativity or research judgment.

No sequencing rule removes the basic risks of AI-assisted learning. Several deserve explicit attention.

The framework adds friction. It is slower than asking for a finished answer. That is partly the point, but excessive structure can become bureaucracy. Instructors should collect the smallest process evidence that reveals thinking and should remove scaffolds as students become more capable.

Novices need more support. A first-year student may not know what counts as a meaningful alternative, a reliable source, or a load-bearing assumption. AI can help, but unguided AI can also hide the gap. Well-designed tutors matter. Bastani and colleagues found that unguarded generative AI improved practice

performance but harmed later unassisted performance, while a more constrained tutor removed the exam penalty without producing a positive exam gain [2]. Kestin and colleagues, in a different course and design, found substantial learning gains from a carefully designed AI tutor relative to one specific active-learning condition [18]. Taken together, these studies argue against treating “AI tutor” as a single intervention. Design details matter.

AI can narrow the space it expands. A model can generate many alternatives while still drawing different students toward common patterns. Cohort-level diversity therefore deserves monitoring in addition to individual creativity [7].

The adversary can flatter. Sycophancy means students should not assume that asking for criticism produces independent criticism. Prompts should specify the adversarial role, and important decisions should encounter human as well as machine opposition [5]. Related evidence from a survey of 319 knowledge workers found that higher confidence in generative AI was associated with less self-reported critical-thinking effort, while higher task-specific self-confidence was associated with more [22]. The study was observational and workplace-based, so it is best read as a caution about calibration rather than a student-learning effect.

Verification can become superficial. Students can learn to perform a verification ritual without testing the parts that matter. Instructors should ask them to identify what would falsify a claim or change a decision, not merely list sources checked.

Creative independence is hard to measure. A strong AI-assisted artifact does not show whether the student became more creative. The most credible evidence comes from later unassisted transfer, delayed reconstruction, or new problem formulation. These measures cost instructional time.

Navigation can become over-optimization. A leverage test is useful when the path is unclear, but creativity also depends on curiosity, play, and occasional inefficient exploration. If students treat every branch as a short-term return-on-effort calculation, they may abandon difficult or anomalous ideas before their value becomes visible. The Next-Move Check should therefore restore direction, not eliminate serendipity.

Domain differences are real. Creativity in mathematics, engineering design, historical interpretation, music, and laboratory science does not look the same. The six disciplines are intentionally abstract. Each field must decide what counts as a meaningful first attempt, valid verification, useful divergence, and convincing transfer.

5.3 Research agenda

The evidence base is promising but incomplete. Several questions would materially improve practice.

- How much independent generation is enough before AI collaboration, and how does the answer vary with expertise?
- Does repeated use of a regulated Learning Spiral produce lasting gains in independent creativity across a semester or year?
- Which AI roles are most useful at different stages: Socratic mentor, adversarial critic, divergent ideator, simulator, or co-author?

- When does AI increase the diversity of an individual's ideas while decreasing diversity across a cohort, and which assignment designs preserve both?
- How should creative transfer be measured in mathematics, science, engineering, writing, and design without reducing creativity to a single score?
- What combination of AI critique, peer critique, and instructor critique produces the strongest calibration without overwhelming the learner?
- How accurately can students learn to identify the current bottleneck and choose informative next moves, and does AI mentoring improve that judgment or merely make the choices appear more sophisticated?
- How quickly can scaffolds be withdrawn while preserving both independent competence and effective AI-assisted Practice?

The most important experiments should distinguish assisted output from later unaided performance. Without that separation, an intervention can look successful because the tool made the product better even if the student learned less.

6 Conclusion

Generative AI gives students access to more explanations, more alternatives, more criticism, and more provisional solutions than any previous study tool. That abundance can support creativity, but abundance alone does not teach a learner how to choose, verify, integrate, or originate.

The most useful educational response is not to keep AI outside creative work. It is to design the collaboration so that the learner repeatedly does the parts of creative work that develop judgment. Generate before consulting. Represent before rejecting. Use AI to widen the search rather than close it. Trace important claims to evidence. Build serious opposition into the process. When the path branches, diagnose the bottleneck and choose one high-value next move. Return to independent work. Transfer the idea to a new problem. End by asking what deeper question the work has opened.

These disciplines are compatible with the Learning Spiral because they are the same learning logic under a more creative objective. The student frames a question, tries first, asks for a particular kind of help, works independently, verifies, reflects, and then carries the result into a deeper question. When several deeper questions compete, the student also practices deciding which one deserves attention next. AI makes each pass richer when it is used as a mentor or collaborator. Human instructors and peers make the pass more trustworthy by adding independent judgment and standards the model does not possess.

The resulting aim is modest but demanding. A course cannot manufacture inventive minds. It can, however, create repeated conditions in which students practice independent generation, encounter unfamiliar alternatives, defend ideas against serious challenge, verify what survives, diagnose where progress is blocked, and learn to recognize the next problem worth pursuing. Generative AI makes those conditions easier to supply at scale. The instructor's task is to ensure that the student's mind remains the place where the final integration happens.

Acknowledgment of source lineage

An earlier companion essay, *Cultivating Inventive Minds*, developed six non-AI disciplines from historical educational traditions. The present paper substantially reorganizes that material around the Learning Spiral and current evidence on human-AI collaboration. The navigation layer generalizes ideas from the author's *WhatToDoNext* (WTDN) protocol, originally developed for identifying the highest-leverage next improvement

in a research manuscript. The video transcript that helped motivate the historical comparison is included in the references [30]; this paper does not rely on that transcript as evidence for the effectiveness of the AI-centered practices developed here. A later RSA Albert Medal conversation with Demis Hassabis contributed practitioner examples of rapid prototyping, orthogonal search, informative failure, and a possible classroom inversion [12]. Those remarks are used as expert perspective and design heuristics, not as efficacy evidence.

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